

Arduino code

Here's a basic **Arduino code for a Dual Axis Solar Tracker** using 4 LDR sensors and 2 servo motors:

```
#include <Servo.h>

Servo horizontal;
Servo vertical;

int ldrTL = A0; // Top Left
int ldrTR = A1; // Top Right
int ldrBL = A2; // Bottom Left
int ldrBR = A3; // Bottom Right

int hPos = 90;
int vPos = 90;

int tolerance = 50;

void setup() {
  horizontal.attach(9); // Horizontal servo
  vertical.attach(10); // Vertical servo

  horizontal.write(hPos);
  vertical.write(vPos);

  delay(1000);
}

void loop() {

  int TL = analogRead(ldrTL);
  int TR = analogRead(ldrTR);
  int BL = analogRead(ldrBL);
  int BR = analogRead(ldrBR);

  int top = (TL + TR) / 2;
  int bottom = (BL + BR) / 2;
  int left = (TL + BL) / 2;
  int right = (TR + BR) / 2;

  int verticalDiff = top - bottom;
  int horizontalDiff = left - right;

  // Vertical movement
  if (abs(verticalDiff) > tolerance) {
    if (top > bottom) {
      vPos--;
    } else {
      vPos++;
    }
  }
```

```

    }

    vPos = constrain(vPos, 0, 180);
    vertical.write(vPos);
}

// Horizontal movement
if (abs(horizontalDiff) > tolerance) {
    if (left > right) {
        hPos++;
    } else {
        hPos--;
    }
}

hPos = constrain(hPos, 0, 180);
horizontal.write(hPos);
}

delay(50);
}

```

Components Required

- Arduino Uno
- 4 × LDR (Light Dependent Resistor)
- 4 × 10kΩ Resistors
- 2 × Servo Motors (SG90/MG995)
- Solar Panel
- Breadboard and Jumper Wires

Connections

- LDRs → A0, A1, A2, A3
- Horizontal Servo Signal → D9
- Vertical Servo Signal → D10
- Servo VCC → 5V (or external power supply)
- Servo GND → GND

This code continuously compares light intensity from four directions and moves the panel toward the brightest light source, enabling dual-axis solar tracking.